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Production Planning & Control: The Management of Operations
Publication details, including instructions for authors and subscription information:
http://www.tandfonline.com/loi/tppc20

Manufacturing execution system - a literature review

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Published online: 28 Sep 2009.

To cite this article: B. Saenz de Ugarte , A. Artiba & R. Pellerin (2009) Manufacturing execution system - a literature review, Production Planning & Control: The Management of Operations, 20:6, 525-539, DOI: 10.1080/09537280902938613

To link to this article: http://dx.doi.org/10.1080/09537280902938613

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Production Planning & Control
Vol. 20, No. 6, September 2009, 525–539

Taylor & Francis
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Manufacturing execution system – a literature review

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(*Final version received 16 March 2009*)

Since their beginning in the mid 1990s, Manufacturing Execution Systems (MES) have significantly evolved into more powerful and more integrated software applications as computing technologies have advanced. Their functionality coverage has changed significantly and can now provide a common and single system to support most of the manufacturing execution processes from the production order release to the delivery of finished goods. However, MES applications still fall short of adding capabilities for decision-making tools in very dynamic organisations where adaptive execution strategies are required to replenish the supply chain while dynamically responding to unpredicted change. This article intends to describe what MES have become, present their relationships with other enterprise information systems, and to identify major issues related to MES. We also discuss research areas that need to be explored in order to resolve the increased complexity of execution systems and to fulfil the continuing customer needs for faster real-time response and expanded functionality coverage.

Keywords: manufacturing; execution system; MES; ERP; adaptive manufacturing

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1. Introduction

For more than 25 years, companies have invested in information systems to achieve productivity gains that have caused the enterprise information system market to grow steadily. Some information systems, such as the Enterprise Resource Planning (ERP) systems, gained a major status in the market, as 80% of the Fortune 500 firms are now using ERP to manage their operations and a growing number of SMEs are adopting the same strategy. But for most of this period, the information system specialists have not paid attention to the shop floor (Holst 2001). In that context, it is not surprising to note that production departments have always favoured the development of custom-made software applications to fill specific necessities as an industrial operations support. Production data collection systems using either databases or spreadsheets are commonly developed on the shop floor to monitor and control real-time and variable execution processes. Software maintenance and data consolidation is obviously complex in such an environment as the number and the structure of these small applications vary over time.

Fortunately, the difficulty of integrating multiple point systems has brought software providers to package multiple execution management components into single and integrated solutions. These systems, commonly referred to as Manufacturing Execution System (MES), provide a common user interface and data management system. The MES concept was born from the demand on the manufacturing enterprise to fulfil the requirements of markets from a point of view of reactivity, quality, respect of standards, reduction in cost and deadlines. As such, the functions of MES are primarily turned towards manufacturing activities, a major vector of the added value of all manufacturing firms. Historically well implemented in process industries (pharmaceutical, food and semiconductors industries) where MES systems answer the needs of traceability imposed by the authorities, they are now used in most manufacturing industries.

This article intends to present the actual state of the commercial MES solutions available on the market and to identify major issues related to their use and implementation. We also discuss research areas that need to be explored in order to resolve the increased

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ISSN 0953–7287 print/ISSN 1366–5871 online
© 2009 Taylor & Francis
DOI: 10.1080/09537280902938613
http://www.informaworld.com

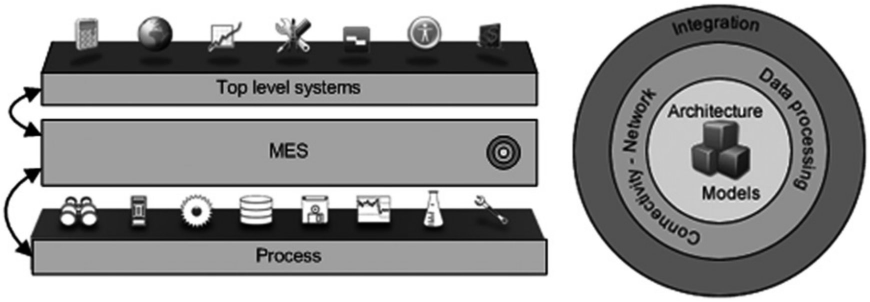

Figure 1. Manufacturing Execution Systems environment and its layers.

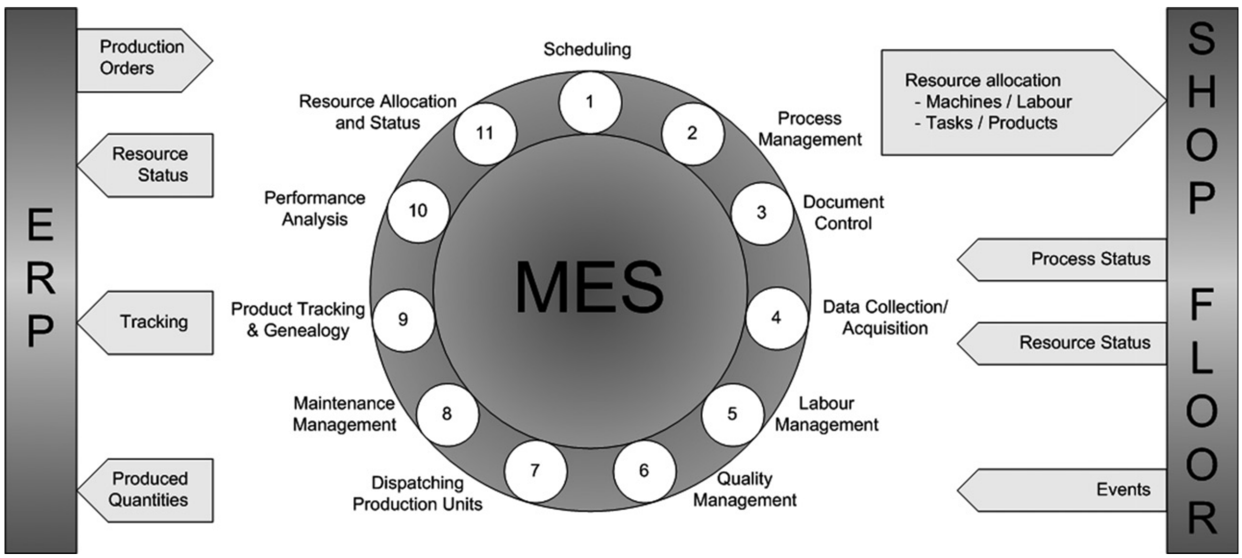

Figure 2. Manufacturing Execution Systems functionalities.

complexity of such systems. The remainder of the article is structured as follows. Section 2 first gives a more detailed description of the MES concept followed by a discussion of the technology trends in Section 3. Sections 4–6 then cover major issues related to MES such as architecture, connectivity and network and data processing. Section 7 presents integration models proposed by the research community. These four sections can be viewed as the core, the nerves and the skin of an MES as represented by Figure 1. Finally, we present the current limitations of MES and discuss research areas that need to be explored to fulfil continuing customer needs for faster real-time response and expanded functionality coverage in Section 8; our concluding remarks are in Section 9.

2. Background

Historically, each software editor had their own definition of an MES which was generally based on the capacities of their own tools or on the expectations of their customers. The MESA organisation made the first step toward the MES standardisation by gathering the major actors of the market and by proposing a formal definition of MES (MESA 1997c):

> MES deliver information that enables the optimisation of production activities from order launch to finished goods. Using current and accurate data, an MES guides, initiates, responds to and reports on plant activities as they occur. The resulting rapid response to changing conditions, coupled with a focus on reducing non-value-added activities, drives effective plant operations and processes. The MES improves the return on operational assets as well as on-time delivery, inventory turns, gross margin and cash-flow performance. An MES provides mission-critical information about production activities across the enterprise and supply chain via bidirectional communications.

To meet the needs of a variety of manufacturing environments, MESA identified 11 principal MES functions (MESA 1997b). They are as follows (Figure 2):

- Operations/Detail Scheduling: sequencing and timing activities for optimised plant

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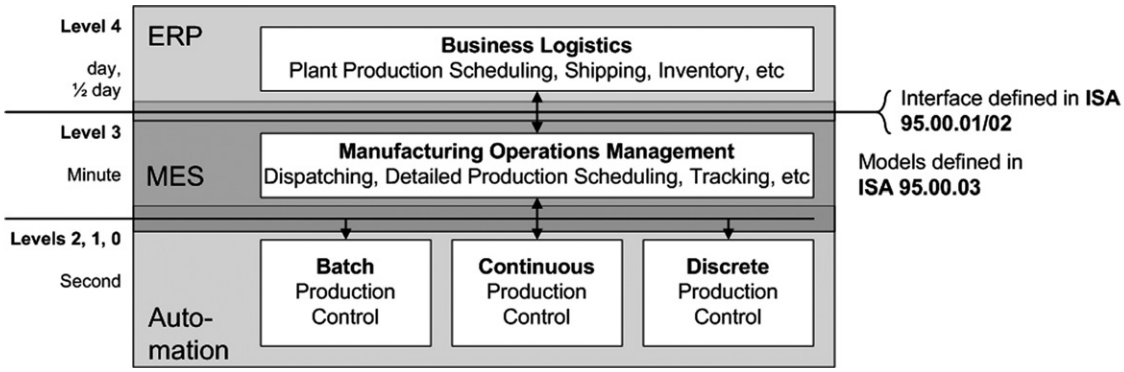

Figure 3. Manufacturing Execution Systems positioning in the Computer Integrated Manufacturing (CIM) context.

performance based on finite capacities of the resources.

- Resource Allocation and Status: guiding what people, machines, tools and materials should do, and tracking what they are currently doing or have just done.
- Dispatching Production Units: giving the command to send materials or orders to certain parts of the plant to begin a process or step.
- Document Control: managing and distributing information on products, processes, designs or orders, as well as gathering certification statements of work and conditions.
- Product Tracking and Genealogy: monitoring the progress of units, batches or lots of output to create a full history of the product.
- Performance Analysis: comparing measured results in the plant with goals and metrics set by the corporation, customers or regulatory bodies.
- Labour Management: tracking and directing the use of personnel during a shift based on qualifications, work patterns and business needs.
- Maintenance Management: planning and executing appropriate activities to keep equipment and other capital assets in the plant performing to goal.
- Process Management: directing the flow of work in the plant based on planned and actual production activities.
- Quality Management: recording, tracking and analysing product and process characteristics against engineering ideals.
- Data Collection/Acquisition: monitoring, gathering and organising data about the processes, materials and operations from people, machines or controls.

Among these 11 functionalities, some are directly linked to the process, such as scheduling and quality control, while others, such as resource management and traceability, are best described as cross functions. This difficulty in classifying MES functions has two consequences. First, the concept of MES is often difficult to perceive clearly by the concerned manufacturers. Second, we tend to bring in everything that is not already assigned to other areas, such as ERP. This latter result does not facilitate its structure. In reality, all MES functions are not on the same level.

The major contribution of the S95 standard, published by the Instrumentation, Systems & Automation (ISA) Committee, is the identification of critical interactions between different parts of an MES system. In the absence of such a system, developed solutions may not be able to follow the changing needs of industry, and may even be unable to effectively respond to the problem. For example, a periodical quality problem may be correctly interpreted if it can be mapped to maintenance operations. Product traceability is inefficient if it is not coupled with the team management on one hand and process control information on the other hand.

The standard S95 deals with the formalisation of exchange around the manufacturing system to other areas of the company. Designed to be applied to all manufacturing strategies, it does not impose any organisational model or manufacturing system architecture. It suggests a physical model for the company extrapolated from the standard S88 and a definition of functions and information flows based on the Purdue University Reference Model (PRM). The function model of the S95 obviously puts the manufacturing control function in a central position. The MES is then an intermediary between the various departments of the company (Figure 3). This standard is now widely accepted by market players for the design of information flows between shop floor level applications and

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those of a higher level. It also allows the industry to have a consistent terminology.

All MES functions contribute to achieving on-time performance and total adherence to customer order schedules. According to MESA's survey (MESA 1997c), MES systems have provided manufacturing enterprises with some of the most impressive benefits of any manufacturing software, such as an average 45% reduction in manufacturing cycle time, a significant improvement of the flexibility to respond to customer demands, the realisation of certain degrees of agile manufacturing and customer satisfaction.

3. Technologies

Manufacturing Exceution System solutions are traditionally designed on a modular basis. Thus, if the MES functions set is theoretically invariable, the priority between them and the configuration can vary and be adapted according to the aims of the company. Some functions are sometimes redundant to those of other systems (mainly with the ERP) but are more focused on the production equipment performances. A good example of this modular concept is InSite™, the Camstar MES (Camstar 2006b), which is composed of 18 integrated modules to collect data and continuously control production activities. In a general way, most MES modules can be classified in three layers (FactoryTalk 2006, Proficy 2006, SIMATIC IT 2006, SiView 2006, Camstar 2006a):

- *Client/server application layer*: This is the layer that is visible to the users. It is also where other systems can be 'plugged in', enabling them to communicate in a distributed messaging environment.
- *Integration framework layer*: The integration framework layer is the heart of the architecture. It is made up of standardised, reusable, object components that provide the application infrastructure through which information flows among applications. This infrastructure is mostly built on standard distributed object tools such as CORBA or COM/DCOM/OLE.
- *Data storage/management layer*: This layer provides essential services for the application and integration framework layers, such as network communications, object management services and persistent object services. These services and facilities must be built on robust, standard operating system and database technologies to provide a solid and long-term foundation for the whole system.

More specifically, the leading MES companies are focusing on various MES enhancement issues, such as integration, scalability and graphical configuration (MESA 1997c). Such improvements are based on the significant advances of network computing (i.e. network protocols, internet, intranet and extranet) and software technologies (object, components, client/server architecture, CORBA, ActiveX, Agent, XML, XSLT, SOAP, web services, universal equipment connectivity and languages). The improvements and the future business characteristic trends can be summarised as follows:

- Integration: the key to the deployment of an enterprise-wide information system, where an MES bridges the shop floor control and office planning systems.
- Heterogeneous knowledge synthesis: mechanisms to make the collected data more understandable and to make the decision making easier for the user.
- Standardised connectivity: a plug-and-play connectivity for equipments suitable for different computing environments.
- Real-time performance: pertinent information such as the status of process, material, resource, personnel and service delivered to the right user at the right time.
- Web-based architecture: open and modular system architecture to improve the interoperability for heterogeneous applications and adaptability for the advances of new technologies.
- Scalability and reconfigurability: a MES must take into account the needs of current companies, their size and their dynamic nature.

4. Architectures and models

Scientific literature has mainly proposed models and architectures for developing a general-purpose and efficient MES. In this section, we review some MES open frameworks to facilitate their integration with the current systems and we also discuss the interaction among multiple MES and the data modelling efforts.

Network technology introduces new possibilities such as client-server architecture, parallel, distributed or concurrent working. It generates important challenges like e-business, B2B or B2C, which go together with global competition. Network applications are currently being developed at all levels of manufacturing engineering, issuing new concepts such as agile manufacturing, virtual enterprises and virtual factories (Jain 1995). Complexity of production increases as

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products are getting more and more diversified and suited to customers' requirements and as manufacturing operations and tools are getting more and more sophisticated. Here, the strongly centralised organisation is a structural limitation for managing the complexity implied by the combinatorial expansion. Multi-agent technology applied to manufacturing systems has received particular attention (Van Dyke Parunak 1998, Shen and Norrie 1999) because of their distributed organisation, high modularity and simplicity of implementation. Agents may represent single resources (a work cell, a machine, a tool), workers involved in a process, etc., as well as products, customers or providers. All together, they perform individual tasks, with interaction among them, in order to fulfil production functions (such as procurement, local planning, task assignment, scheduling, execution control, dispatching, etc.). One of the most fruitful and widespread approaches formalising the use of agents in manufacturing systems has been developed in the Holonic Manufacturing Systems (Valckenaers *et al.* 1994), where a 'holon' is defined as the association of a software agent with a physical device or a set of physical devices. Although the cooperation of holons does not use either global or centralised procedures, their particular architecture maintains a hierarchical structure based on a functional decomposition. Multi-agent architectures usually use hierarchical and heterarchical structures (Dilts *et al.* 1991, Leitão and Restivo 1999). In a hierarchical architecture, there are multiple levels of master–slave agent type relationships, whereas in a heterarchical architecture, agents communicate as 'peer-to-peer' without a predefined master–slave relationship. Agents are locally autonomous and cooperate in a negotiation procedure in order to reach their local objective. The advantages expected from heterarchical structures are their reduced complexity, enhanced modularity and scalability, and their intrinsic robustness and fault tolerance. Decisions are made locally only when needed.

The PABADIS project (Plant Automation BAsed on DIstributed Systems) (PABADIS Group 2001, Diep *et al.* 2003) aims at developing an MES based on a networked flexible manufacturing system, designed according to a distributed and de-centralised framework and implemented with autonomous agents. It addresses flexible manufacturing units with small production batches, and will enable future plants to be highly reactive to a turbulent market. Holons are also used in a MES architecture dedicated to mass customised manufacturing (Simao *et al.* 2006). SEMATECH (SEMATECH Inc. 1998) is another open framework for the integration of MES running in semiconductor industries and focused on the control at the shop level of manufacturing systems. It incorporates basic manufacturing functionalities at the factory engineering level, which include product design, process planning and material requirement planning (Harhalakis *et al.* 1994, 1995). It comes with analysis and design tools based on UML diagrams and Petri nets (Lin *et al.* 2005, Lin and Jeng 2006). In the same frame of thought, we can talk about NIIIP-SMART (Barry *et al.* 1998) architecture design with a standards-oriented configurable object model that represents the diverse aspects of MES in order to reduce design costs, or OpenMES (Hori *et al.* 1999) which is an object-oriented framework with a modular organisation to achieve open connectivity not only with other software systems, but also with manufacturing devices.

Most MES research papers report the construction of a single MES and seldom consider its interactions with multiple MESs that represent multiple shop floors. Now, collaborative business has become common and outsourcing functions in enterprises gradually play an important role. Terms such as extended enterprises, contracting organisation and virtual enterprises have been coined for investigating the special cooperative behaviours among companies. However, research usually focuses on the study of value chains and strategic alliances. Little research is done on the execution level. Huang (2002) proposes a distributed workflow model based on cooperation types and operation workflow. But there are major limitations because of the different data models used in the distributed MES.

Since the database is the core in an information system, modelling efforts are not only focused on the architectural aspect. The key steps to designing, building and implementing a cost-effective, integrated MES to plant needs are to assess the present state of integration and the availability of infrastructure and then to map the information needs of all the operational layers. Based on this, the database of the MES has to be designed for the plant-specific conditions. Although the third part of the S95 standard attempts to describe the data model of MES, some authors do not await its publication to make proposals. For example, Zhou and co-workers (Zhou *et al.* 2005) adopt the extended entity-relationship model technique to build the data model of MES.

5. Connectivity and network

The demand for automation of manufacturing processes is rather high, and most manufacturers make efforts to adopt and/or improve the technique of equipment integration and communications. Yang *et al.* (1999), Cheng *et al.* (2000b) have developed a generic equipment manager (GEM) which can either operate alone or be integrated with the MES. In order to achieve generality and reusability, the governing rules of the GEM are divided into three independent portions: System/Framework Rules, Business Rules and Equipment Rules. In the first release, these rules were implemented in the MES interface and equipment driver modules separately. Among them, equipment rules in the driver modules are equipment-dependent. Connecting a different machine to the equipment manager requires modifying the associated driver. To solve this problem, the authors have proposed a web-enabled equipment driver that uses mobile object technology. The equipment manager downloads the driver via a web browser, and subsequently, it can communicate with the equipment by either the Java socket technology or the object web technology through the CORBA or DCOM protocols (Cheng *et al.* 2000a).

Ye and Qiu (2003) and Qiu (2003) define generic equipment connectivity as a platform capable for continuous customisation. By loading configurable equipment connectivity components and an equipment characteristic descriptor, generic equipment connectivity is then enabled to support proper communication protocols. A schema based on the equipment description is used as an effective vehicle for assimilating different information structures and semantics and to extract pertinent information among overwhelming data from heterogeneous sources.

Networked controlled systems should integrate all new technologies such as wireless networks, embedded systems, nomad components and electronic tags (RFID), to meet new requirements such as mobility, modularity, control and diagnosis decentralisation or distribution, autonomy and redundancy, allowing quick and easy maintenance. A major industrial communication challenge of the related multilevel communication architectures is to unify plant networking with Ethernet. The resulting challenge is to guarantee the same deterministic features as those of more specific fieldbuses currently applied in a shop floor. For example, applying fault detection and isolation/fault tolerant control techniques to networked control systems should improve safe control and monitoring of such complex systems as well as their global reliability, dependability and availability by dynamically accommodating network performance, reconfiguring network components and adapting the application to the delivered quality of service (Georges *et al.* 2006).

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6. Data processing

As mentioned previously, data and all that surrounds them are the core of the effectiveness of an MES system. It needs to analyse and put data into context for the user. Like information flows before material does and if the information within a manufacturing enterprise is slow and of poor quality, how can the quality and high-velocity production be ensured? It is clear that having the right information at the right time helps to make the right decisions. Current technology can ensure the availability of information, but it cannot ensure that the information is accurate and meaningful (Jones *et al.* 2002). If the precision of acquired information depends on the chain of acquisition implemented, the meaning associated to this information only depends on the analysis that we make (Slater 2000, Rockwell 2006).

One possibility for gathering data for analysis is to extract data directly from one or more operational databases. Operational or transactional databases are designed to process individual transactions quickly and efficiently. This type of transaction-based interaction is known as on-line transactional processing (OLTP). When transactional data is no longer of value to the operational environment, it can be stored in a data warehouse to be used by a decision support environment. An entity-relationship model is often adopted to create the data models for OLTP system. However, multi-dimensional data models with a star or snowflake (Wrembel and Koncilia 2006) are adopted as modelling techniques for a decision-based data warehouse. On-line analytical processing (OLAP) is a query-based methodology that supports data analysis in a multidimensional environment. An OLAP engine logically structures multidimensional data in the form of a cube. MS Excel provides an interface that allows us to view OLAP cubes. Thus, OLAP cubes and MES data warehouse can be used to analyse shop floor data. Chen and Wu (2005) design OLAP cubes for analysing equipment performance, WIP and quality.

On the shop floor, data collected from equipment, devices and other instrumentations are normally at the necessary level of granularity for fine-tuning manufacturing processes. For instance, they are widely used for statistical process control and run-to-run process adjustments (Xu and Qiu 2004). However, if a higher office level information system needs to use this information, they will be most likely of no value to a user without being filtered and put into the context of the relevant business unit domain (Baliga 1998, Slater 2000, Qiu *et al.* 2002). To filter data, standardised data processing technologies, such as document object model (DOM) can be used (Qiu 2002, Ye and

Qiu 2003). The extraction rules are application/module-dependent but independent of various data sources. Using XML technologies (XML Core Working Group 2003) makes the integration dynamic and extensible. In addition, combining XSLT language (XSL Working Group 2001) and XML may permit to dynamically modify the data object configuration (Xu and Qiu 2004).

7. Integration models

Manufacturing Execution Systems have been researched and developed to narrow the gap between the shop floor and the office-planning systems (Owen and Parker 1999, Slater 2000, Koch 2001). Obviously system integration to make shop floor data (e.g. inventory availability, equipment utilisation, supplier's status, customer orders, commitments and production schedules) available in a timely way is fundamental to the realisation of an integrated enterprise-wide information system, where effective connectivity to the shop floor is not an option but the cornerstone for the well-integrated enterprise-wide information system (MESA 2000). The benefit of such an integration will be different for each company but the common factors are visibility of processes, ability to make them more responsive and provision of track and trace capabilities. All manufacturing organisations have an integration solution but in most cases they are paper- or people-based (Adshead 2007). We can split the integration approach into two major categories: the software integration – thanks to application programming interface (API) or agents – and data integration.

Currently, most office information systems are not designed to work at the shop floor level. Howells (2000) observed that these systems began as financial applications, moved into human resource, logistics, marketing and supply chain applications, and often as an afterthought, attempted to link to shop floor manufacturing systems. Lacking a comprehensive design-for-integration consideration in the design stage, office information systems can hardly be connected to the shop floor. When there is a need, the point-to-point API approach has consequently been adopted by most manufacturing enterprises, giving rise to extremely costly, non-flexible, non-scalable and hardly sustainable system architecture (Linthicum 1999, Slater 2000, Koch 2001).

After the development of large-scale, non-open and owner solutions, the interfaces' definitions such as the Business Programming Application Interfaces (BAPI) of SAP R/3 became a way of integration in the heterogeneous systems. Because of the need to combine the best modules of various vendors, the concept of reusable preset components was developed and applied to integrate various systems within the same company or between organisations to develop their coordination (Mustafa and Mejabi 1999). With this kind of interface, it is possible to be integrated in the internal processes of the software. For example, the maintenance models – which are the subject of much research – are often integrated into the ERP. Ip *et al.* (2000) use the integrated definition method (IDEF) model to systematically integrate maintenance into MRPII. Whereas Nikolopoulos *et al.* (2003) have preferred the object approach, Rizzi and Zamboni (1999) developed a module integrated into an ERP to optimise the perishable goods storage.

As in the PABADIS project, mobile agents can also be used to model MES architectures. But the capacities of agents to negotiate in an autonomous way to succeed in achieving the goal which is assigned to them can also be used as integration vector. They were already implemented to manage the capacity of a shop floor from an ERP (Mertens *et al.* 1994) or to incorporate shop-floor controls formally into plant-wide information-control systems for enabling 'on the fly' rescheduling of product routes as well as manufacturing process reconfigurability (Tang and Qiu 2004). The development of this method is made easier by the JAVA programming language capabilities, which natively implement the mobile agents and are able to transform the complex architecture of an agent into simple text file and vice-versa (Wong *et al.* 1999).

Howells (2000) also pointed out that the reality of manufacturing businesses was based on the facts that:

- Regardless of the products manufactured, a financial package can be implemented in any industry.
- Regardless of the products manufactured, most logistics requirements are similar.
- It is in the manufacturing plant that the unique differentiators and business data exist.

In other words, different manufacturing equipments (i.e. processes) are used in making different products, such as automobiles, food, chemicals, etc. It is the variety of types of manufacturing equipment that complicates the shop floor connectivity. At the present time, data are transported from various sources in various formats using various communication interfaces. Efficient data integration will be best suitable for the integration architecture. For example, Wong (2006) proposes a data model to combine an ERP and a PLM. With S95 standard, the goal of shop floor to 'top floor' integration is easier to achieve. Integration has been

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done for many years with CVS files and spreadsheet macros resulting in non-flexible integration. Now, things are relatively standard to connect the shop floor with MES and ERP systems. For example, OPC (OLE for Process Control) can gather data from control systems and B2MML (the XML implementation of S95 standard) and get it up to ERP application (Adshead 2007). XML standard is the ideal approach to data harmony: the integration becomes dynamic, extensible and can be dynamically modified (Salam *et al.* 1999).

Lobecke and Slawinski (2004) also discuss the integration of an MES and an ERP. It turns out that a consistent data model is essential for the horizontal and vertical integration of the systems.

8. Limitations of current MES and challenges

Limitations referred to here mainly concern the practical environment where the MES operates. We discuss the three layers represented in Figure 1: the *architectural* aspect, the *connectivity* impact on the real-time data processing and the *integration* difficulties of such systems. We also evoke some tracks proposed in the literature to reduce the impact of those limitations and some challenges for MES evolution.

8.1. *Architecture*

Today, the major architectural issue is the lack of tools and platforms to test and validate new architectural developments on realistic problems, in terms of both the size of the manufacturing system and the thoroughness of the evaluation techniques. Recently, research on applying multi-agent systems in manufacturing has produced many results (Muhl *et al.* 2003). However, various obstacles remain for deployment in industry. Marik and Lazansky (2007) emphasise that only very few real-life industrial experiments are in effect, despite laboratory experiments on the promising MAS and HMS approaches. Therefore, an environment is required in which the research community can provide and retrieve test cases of realistic size and complexity; in other words, research developments must be tested in real-world emulated factories in place of the toy test cases and token evaluation campaigns which remain the norm.

8.2. *Real-time processing*

Theoretically, an MES must be able to process the data of the shop floor in real time (minutes/hours) to be able to control the manufacturing activities (MESA 1997c, 1997b). To be reactive, it must not only obtain these data, but also analyse them in real time. When controls are made in seconds or fractions of seconds in shop floor process execution systems, the MES may take minutes or even hours to respond. Compared with the chaos of old shop floors, where no current and accurate shop floor information was available and ERP was implemented as open-loop systems with response times in days, the current MES solutions are relatively 'real time'.

In general, the scalability of a current MES is very much limited in terms of the capability to keep its real-time resolutions when the governed manufacturing system grows. It is evident that the amount of information collected from control systems increases tremendously with the degree of increased automation on the shop floor. That is to say, MES are required for processing more information. Manufacturing systems grow because of the need for more complex processes to meet the needs of increased product functionality. In other words, MES are required to be either physically or logically connected to more equipment and are dealing with tremendously more data. It is a vicious circle: the more an MES wants to be closest to the real-time notion, the more it must be connected; the more it wants to be connected, the more it wants to process data; finally, the more it wants to process data, the less it can be closest to the notion of real-time processing.

In addition to being able to improve the productivity of a global enterprise manufacturing system, the deployed MES in such a circumstance has to be capable of obtaining all of the manufacturing process and product data from all plants geographically dispersed around the world in a timely way. These data should also be analysed in real time to make enterprise-wide optimal decisions such as validating the equipment setup and production schedule, monitoring the process status and indicating the trend of product quality, reacting more quickly to dynamic customer demands when parts of a typical order could be simultaneously made in different plants.

8.3. *Integration*

First of all, establishing a hierarchy of production tasks, which has always been the unique organisational scheme, seems to be supplanted by other forms of management. In conventional systems like MRPII, MES executes and controls production orders which emanate from ERP. This top-down structure does not easily integrate different production forms, such as Just-in-Time (JIT) production, 'pull' production, or inverse manufacturing which favours a bottom-up production demand (Massotte 1997), and necessitates a change in the processing of orders.

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Another difficulty comes from the software heterogeneity of the manufacturing environment. Indeed, it is not rare within a same MES solution to have to implement modules of various providers, each one with its data model, its communication mechanisms and even its own data base. To have them work as a whole, proprietary mechanisms are commonly applied for system integration. Furthermore, because MES applications are often mission-critical while lacking proper adaptability to new technologies, they may not be replaced or updated as the technology advances.

The adoption of owner formats and API by the providers stays, and shall probably stay, the major reason limiting the integration of these tools. It is why the implementation, the integration and the maintenance of MES are so expensive. The large companies miss time and the small ones miss resources to conclude these projects (Koch 2001). Although the MES development is a prosperous industry, the high costs of development and the owner formats slow down the adoption of these tools in many companies. By nature and definition MES are ready to be integrated. The problem is that each information system (ERP, SCM, etc.) has owner interfaces, and in particular the manufacturing equipment is innumerable and quite often unique. Academic research on integration problems show it and all are unanimous in stressing the importance of the integration of MES with other information systems. Since 1996, Westerlund (1996) has insisted on the importance of the integration capacities in the choice of ERP and MES systems because it is one of the keys to achieving financial goals. In 2001, Rondeau and Literal (2001) reiterated this call. Liu *et al.* (2002) describe the difficulties in integrating an Advanced Planning System (APS), an ERP and an MES to check the capacity constraints of the production plans.

The Service-Oriented Architecture (SOA) technology can solve those integration issues. SOA is a design for linking computational resources (principally, applications and data) on demand to achieve the desired results for service consumers (who can be end users or other services). This style of architecture promotes reuse at the macro (service) level rather than the micro levels (e.g. objects). It can also simplify interconnection to and usage of existing IT (legacy) assets. SOA is an architectural style whose goal is to achieve loose coupling among interacting software agents. A service is a unit of work done by a service provider to achieve desired end results for a service consumer. Both provider and consumer are roles played by software agents on behalf of their owners. The only disadvantage of this solution is that it would require the complete rewriting of all the current applications (Erl 2005). But nowadays, many researches and major contributions have been found on service-driven manufacturing platforms (Chazalet and Lalanda 2007, Jiang *et al.* 2007).

Manufacturing execution is a complex task because of the non-linear nature of the production system, the production processes and the environmental uncertainties. Schedules and plans, originating from higher levels in a manufacturing organisation, can become ineffectual within minutes on a shop floor. Moreover, the variety of existing manufacturing system types and performance issues, as well as the different kinds of equipment and processes, is very wide. To cope with this heterogeneity challenge, future MES designs must apply the most fundamental and recent insights in self-organising systems, a topic that is being intensely investigated by the multi-agent systems community today (Di Marzo *et al.* 2004). To design such self-organising systems, it is also essential to apply insights from fundamental research (Waldrop 1993, Valckenaers *et al.* 2003) and to define the related modelling framework to obtain the required system features. Important expected progress in the domain is the emergence of MESs that are able to forecast the emerging state of the manufacturing system while preserving the level of decoupling that has made older multi-agent MES robust and configurable (Valckenaers *et al.* 2004).

8.4. *Research challenges*

For modern companies, processes are the privileged form of action, they are the added value. It is also why they have invested in the integration of their processes into ERP. But until the emergence of new requirements in reactivity, in quality and in adaptability, the manufacturing processes were forsaken. The objective behind the MES concept is the optimisation of the manufacturing processes and resources. The first step is, naturally, the performance measurement of the current system. MES is naturally strongly connected to the manufacturing processes. It is the best tool to measure in real time performance indicators such as the use of materials, the productivity according to batches, and the machine breakdown. Then, it is possible to create piloting dashboard by evaluating indicators such as the synthetic yield, the mean time between failures or the mean time to repair. Consequently, the actions to be set up appear obvious. Here we find the steps of the continuous improvement diagram: Measure, Analyse and Improve. Those steps largely use data-processing technologies, but also require human expertise. It is a first step in the decision-making support: the MES alarms, presents

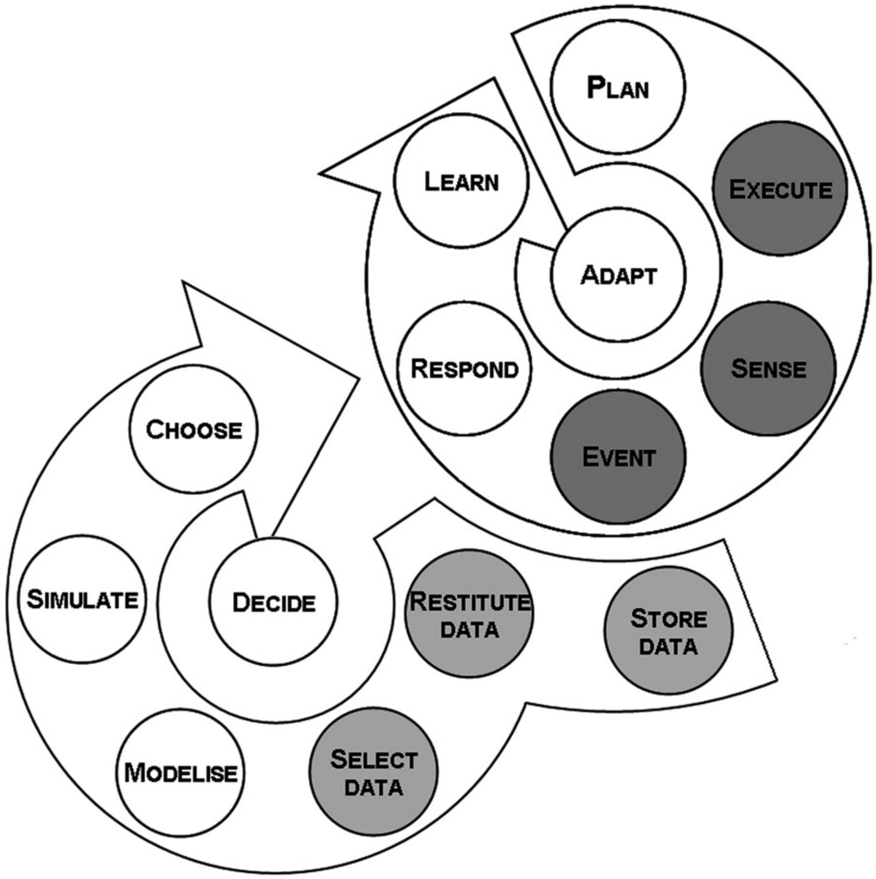

Figure 4. Adaptive intelligent manufacturing system. In grey: covered by current MES.

and formats the data wished for by the user. But the decision support tools fall not only into information processing but also into operational research and analytical methods. The contribution of those systems can result in intervention to one of six possible levels in the decision-making process (Decide loop of Figure 4): (1) elementary data storage, (2) the data return, (3) the relevant data selection, (4) the decision model design, (5) the decision evaluation and (6) the selection and choice of a solution.

Lobecke and Slawinski (2004) present the integration of an MES and an ERP. It turns out that a consistent data model is essential for the horizontal and vertical integration of the systems. When the business processes are well integrated in the data model, optimisation potential can be exploited. The MES functionality implements a set of optimisation methods that rely on a fixed, rigid plant model. Actually, most processes are not or cannot be explicitly modelled, which leads to the occurrence of unexpected events (Homem-De-Mello *et al.* 1999). The MES integration with the other information systems will make it possible for organisations to quickly respond to disturbing events that arise on the shop floor.

Manufacturing Execution System applications do not address all of the manufacturing execution processes required to replenish the supply chain while dynamically responding to unpredicted change. But the MES integration with other information systems will make it possible to adopt adaptive manufacturing strategies by quickly answering to disturbing events appearing in the shop floor. Figure 4 presents the adaptive manufacturing strategy loop combined with six levels of the decision-making process where grey activities can be supported by MES. While MES is seen as the cornerstone of such approach, it still needs to be combined to SCM, PLM and other information system functions which reinforce the need for adopting an SOA or multi-agent design approach. Also, adaptive manufacturing strategies require defining the process management layer between ERP and process control and translating demand-driven supply chain requirements into a set of capabilities, systems and workflow integration investments (Martin 2004). Therefore, adaptive manufacturing organisations need an execution platform that connects manufacturing processes with enterprise and supply chain processes, enables closed-loop mechanisms, and provides decision support

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to production personnel so they can deliver on their performance goals.

In addition, unexplored research issues remain to develop the decision-making support capacities:

- one difficulty is to develop explicative models in order to react in real-time to hazards like defects or failures, and to integrate stochastic treatments;
- the cooperation between high level planners and schedulers and the MES is virtually unexplored;
- scaling the MES technology to multi-site manufacturing coordination and control is only in the initial stages of research.

8.5. *Industrial challenges and opportunities*

As recent surveys (MESA 1997a, Fraser 2004) revealed, firms measuring and linking operations and financial performance indicators perform best. Also, surveyed businesses using automated data collection on the shop floor achieved better improvements in quality, throughput, customer service, compliance, asset utilisation and inventory. All these arguments should motivate manufacturers to adopt MES solutions. However, justifying the implementation of MES to improve manufacturing and financial performance is a difficult task for many executives. As MES implementation can bring intangible benefits and involves significant costs and risks, a comprehensive and systematic assessment is necessary before making such a critical investment decision.

Traditionally, system investment projects are evaluated based on financial criteria, such as return on investment, net present value or internal rate of return. This approach may underestimate the value of MES as it considers only tangible and monetary effects and ignores intangible ones such as flexibility and agility. Fortunately, Liang and Li (2007) propose a decision methodology based on *Analytic Network Process* to render it facile for a manufacturer to make the decision of whether or not to implement an MES and which solution should be chosen. However, other questions still need to be answered: Which processes should be implemented first? Which data should be collected? Differences in priorities, manufacturing processes and enterprise environment will certainly lead to different decisions from one organisation to another. In all cases, manufacturers should favour processes and data which will generate meaningful performance indicators. Indeed, MES provides visibility to accurate, high-velocity information about production performance at a fraction of the cost and time of an ERP initiative.

An MES implementation project can also become an opportunity to upgrade processes, to recognise and then seize new opportunities both internally and in the marketplace. To achieve this, shop floor staff need more than data analysis. They need decision tools to dynamically respond to unpredicted change. MES is the appropriate application not only to gain access to critical data in real time but also to propagate the decision made. To do so, MES solutions need to combine different solving tools with multiple data sources to elaborate solutions in reasonable time compatible with the planning horizon. The recent advances in computing power and memory have opened up the possibility of on-line simulation-based optimisation. This approach offers one of the most interesting opportunities in simulation and the potential benefits in this field are significant and are just beginning to be exploited. A typical area of application for online simulation, also called real-time simulation, is proactive decision support for scheduling problems in manufacturing systems (Siemiatkowski and Przybylski 2006). For example, deployment of a decision support system in a plant of Nippon Kokan Co. Ltd. allows to divide by two the metal waiting times, which resulted in a reduction of 1 million dollars a year in production costs (Numao 1994). Saenz de Ugarte *et al.* (2006) proposed a real-time execution architecture adding a decision module to the ERP-MES tandem. This architecture is based on efforts already deployed to integrate MES with ERP, and provides a demonstrative example based on a real manufacturing scenario in the aluminium industry.

9. Conclusions

This article presented a literature review of MES which had been developed some years ago to fill a gap between shop floor information and control systems and other enterprise information systems. It presented different approaches proposed by the scientific community in order to facilitate its implementation and integration in complex production environment. We also discussed the limitations of the current MES and the challenges for the future of this system.

Despite all the advances in this area, we note that this industry and the research community have mostly addressed the manufacturing execution problem from a systemic point of view. However, if we want manufacturers to move from a 'react to forecast changes' mindset to an 'adaptive' mindset, researchers must support them by developing new models and approaches which will proactively detect production exceptions and provide production personnel with on-

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line decision support with the objective of bringing the factory and its supply chain to react to the moving targets imposed by its customers.

Indeed, potential research areas are numerous, varying from the development of new re-scheduling algorithms to the development of adaptive performance indicators. On-line decision making tools are important and should integrate evaluation methods. Simulation remains a powerful tool for evaluating different scenarios and calculating performance indicators for every considered strategy. The combination of on-line simulation and optimisation algorithms (mainly heuristics) is now quite established in the academic world but a lot still remains to be achieved in industry. Data mining tools would also help in explaining information gathered at different levels. Aggregating efficiently data to have consistent and significant global performance indicators and their interaction with lower level indicators continues to be an interesting research topic.

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